

Quality Management

2 — Standards, quality, certification and liability

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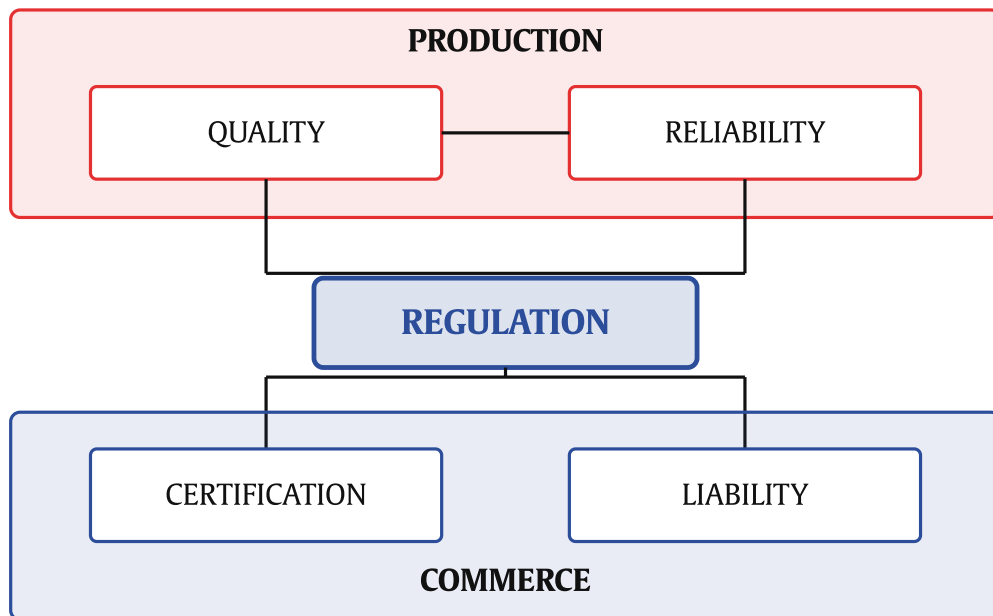
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Where we were

The map, from last time



Last time we did the outer frame: the single market, the barriers it had to remove, and the New Approach that split law from standards.

We left one thing unexplained. The law points at *harmonised standards* — but who writes them, and why does anyone obey a document that is not law?

Today

Standards • quality • reliability •
certification • liability

Today's route

Four of the five boxes on that map

1

Standards

who writes them, how, and why they are obeyed

2

Quality & reliability

two of the four key concepts

3

Certification

who says you conform

4

Liability

who pays when you do not

The thread

Each of these exists because the one before it is not enough on its own. A standard with nobody to check it is advice. Certification with nobody to check the certifier is a logo. And all of it would be optional were it not for liability at the end.

PART 1

Standards

*Who writes the rules that the law points at, and why
anyone follows them*

What standardization is

Definition

The activity carried out to establish, in relation to actual and potential issues, provisions for common and repeated use, aimed at achieving the best order in a given context.

Standardization has grown from *dimensional unification* — making parts interchangeable — to defining the performance of products and processes. That brought two things:

- **Greater effectiveness** — it now influences every stage of a product's life, from concept to disposal.
- **Greater dynamism** — it follows technological change closely, rather than lagging a decade behind it.

KEEP THIS STRAIGHT

A **technical standard** is voluntary. A **technical rule** (*regola tecnica*) is mandatory, issued by a public authority. The New Approach works precisely by keeping these two separate.

Why standards matter

The socio-economic case

- **They make production more efficient** — by unifying products and processes, and by fixing performance criteria and methods for control, testing and inspection.
- **They make technical communication possible** — standardised symbols, codes and interfaces mean a drawing means the same thing in two companies.
- **They protect people and the environment** — by defining requirements for products, processes and behaviour.
- **They safeguard consumers** and the wider community.

A STANDARD YOU CAN SEE FROM SPACE

A shipping container is standardised in eight dimensions. That single agreement is what lets a crane in Shanghai, a ship, a train in Trieste and a lorry in Perugia all handle the same box without anyone phoning ahead. Before it, loading a ship took days of manual labour; after it, hours.

Standards you used before breakfast

All invisible, all load-bearing

- **The plug in the wall** — pin spacing, voltage, insulation class
- **The A4 sheet** — ISO 216, which is why the paper fits the printer and the envelope
- **The screw thread** on anything you could disassemble
- **The date format** in your files — ISO 8601
- **The USB-C connector** — and the EU decision to mandate it
- **Wi-Fi** — IEEE 802.11, or your laptop would speak only to its own brand of router
- **The shipping pallet** under everything in the supermarket
- **The coffee itself** — graded, sampled and tested to method standards

NOTICE WHAT THEY HAVE IN COMMON

You have never read any of these documents, and you would notice within seconds if any of them were withdrawn. Standards are most visible in their absence — which is exactly why they are politically easy to underfund.

Who writes standards

Standardization bodies are recognised organisations — generally associations — that do this continuously. Their job is to bring every interested party into the drafting and to publish the result correctly.

They are meeting points for three groups whose interests do not naturally align:

Producers

want it achievable and cheap

Consumers

want it safe and comparable

Public administrations

want it enforceable

The output

Technical standards are technical specifications whose application is **recommended, not mandatory** — until a law or a contract points at them, at which moment they become binding in practice.

How a standard actually gets written

Years, not weeks — and that is the point

1

Proposal

a need is identified — by industry, a regulator, or the Commission

2

Working group

technical experts nominated by member bodies draft the text

3

Public enquiry

the draft is circulated; anyone may comment, and comments must be answered

4

Vote

national member bodies vote — one country, one vote

5

Publication

and periodic review thereafter, typically every five years

WHY SO SLOW

Because the legitimacy comes from the process, not the content. A technically excellent document produced in three months by four companies would not be a standard — it would be those four companies' preference. The delay *is* the consensus.

What makes a standard legitimate

Five conditions — a document failing any of them is not a standard

- **Public** — accessible to anyone, so that its existence and content can be known.
- **Consensual** — developed with the cooperation and general approval of all interested parties, so it does not serve one interest.
- **State of the art** — based on the combined results of science, technology and experience at a given time.
- **In the common interest** — aimed at the benefit of the community as a whole.
- **Approved** — adopted by a qualified and recognised body at national, regional or international level.

WHY CONSENSUS IS THE HARD ONE

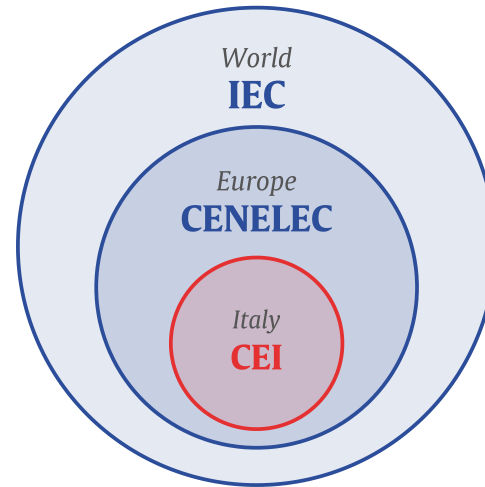
A standard written only by the three largest manufacturers in a sector would be technically excellent and commercially devastating for everyone else. The consensus requirement is what stops standardization becoming a cartel.

The universe of standards

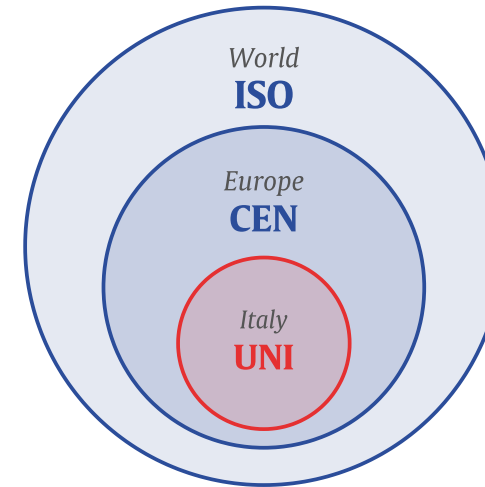
Three levels, two families



Electrotechnical



All other sectors



Telecommunications is the third European body: **ETSI**.

Who is who

ISO — International Organization for Standardization

Founded in London in 1947. Covers all sectors except electrical and electronic. Italy is represented by UNI. Tens of thousands of technicians contribute each year, and it has published on the order of 10,000 standards.

IEC

The International Electrotechnical Commission — the electrical and electronic counterpart to ISO.

CEN — Comité Européen de Normalisation

Founded 1961, alongside CENELEC for the electrical sector and ETSI for telecommunications. Promotes the use of international standards and harmonises them across Europe, removing obstacles created by differing technical requirements.

UNI — Ente Italiano di Normazione

Represents Italy: publishes and disseminates standards, maintains relations with counterpart bodies worldwide, and works with the ministries.

How to read a standard reference

The prefixes tell you the whole journey

UNI EN ISO 9001 :2015

- **UNI** — adopted as an Italian national standard
- **EN** — adopted as a European standard. Member states must adopt it identically and withdraw any conflicting national standard.
- **ISO** — originated as an international standard
- **9001** — the number • **2015** — the edition year

WHAT THE ABSENCE OF A PREFIX TELLS YOU

UNI 9910 — the reliability standard we meet shortly — has no EN and no ISO. It is purely Italian: never adopted at European or international level. **EN 1090** has no ISO: a European standard with no international parent. The prefixes are a compressed history of who agreed with whom.

- **1 A standard is voluntary; a technical rule is mandatory.** Keeping them separate is what makes the New Approach work.
- **2 Legitimacy comes from process, not content** — public, consensual, state of the art, in the common interest, approved by a recognised body.
- **3 Three levels, two families:** IEC→CENELEC→CEI for electrotechnical, ISO→CEN→UNI for everything else, ETSI for telecoms.
- **4 The prefixes are the history.** UNI EN ISO 9001:2015 means international, then European, then Italian — 2015 edition.

PART 2

Quality and reliability

*Two of the four concepts the standards system exists
to serve*

Quality assurance

It is about the organisation, not only the product

Quality assurance covers not just what comes off the line, but the organisation that produced it. You cannot inspect quality into a product at the end; you build the system that produces it.

Its goals are commercial as much as technical:

Establish the product in the market

Improve the economic efficiency of the company

Streamline production

Reduce post-sale interventions

WHY THE LAST ONE PAYS FOR THE REST

A defect caught at design costs almost nothing. Caught on the line it costs a part. Caught by the customer it costs the part, the shipping, the engineer's time, and the next order.

Where standards enter quality

Technical standards do the work in three places:

- **Contractual relations with suppliers** — naming a standard is how you specify what you are buying without writing a specification from scratch.
- **Defining product requirements**
- **Training management and the workforce**

Quality systems then address two quite different situations:

Non-contractual

The supplier's organisation voluntarily establishes and maintains a quality system — because it is worth doing.

Contractual

The client *requires* the supplier to demonstrate quality assurance. This is where certification stops being optional.

NAMING A STANDARD IS A COMPRESSION TRICK

“Bolts to ISO 898-1, property class 8.8” is six words. Writing out the chemical composition, tensile strength, proof load and test methods it stands for would run to pages — and every page is a chance to introduce an ambiguity you will argue about later.

Reliability

Three things at once

- **A property** — the ability to perform the required functions under specified conditions and for a predetermined period.
- **A discipline** — the body of methods that predicts and controls how objects behave over time.
- **A set of activities** — technical and organisational, running through design, manufacture and service.

UNI 9910

Defines the terminology of reliability and the related mathematical quantities — failure rate, MTBF, availability. We use these in the reliability sessions.

THE THREE WORDS THAT DO THE WORK

Required function, specified conditions, stated period. Remove any one and “this component is reliable” means nothing at all. A bearing rated for 20,000 hours at 3,000 rpm and 80 °C tells you something. “A reliable bearing” tells you nothing you can buy on.

- **1 Quality assurance acts on the organisation**, not only the output. You cannot inspect quality in at the end.
- **2 Standards do three jobs in quality**: they specify what you buy, define product requirements, and train people — and naming one compresses pages of specification into a line.
- **3 Quality systems are contractual or non-contractual**. When the client requires proof, certification stops being optional.
- **4 Reliability needs three qualifiers** — required function, specified conditions, stated period. Drop one and the claim is empty.

PART 3

Certification and liability

*Who says your product conforms — and who pays
when it does not*

Certification of conformity

Definition

The act by which an **independent third party** declares, with reasonable assurance, that a specific product, process or service conforms to a particular standard or other normative document.

Two words carry the whole idea:

- **Third party** — not the producer (first party), not the customer (second party). Someone with no stake in the answer.
- **Reasonable assurance** — not certainty. Certification reduces risk; it does not eliminate it.

CONTRAST WITH THE CE MARKING

The CE marking is a *first-party* declaration — the manufacturer's own. Certification is third-party. This is exactly why the CE marking is not a quality mark.

Three different things can be certified

They are routinely confused, including in job adverts

Product certification

This *product type* conforms to this standard. Tested against product characteristics — a helmet, a cable, a valve.

System certification

This *organisation* runs a management system meeting a standard — ISO 9001 for quality, ISO 14001 for environment. Says nothing directly about the product.

Personnel certification

This *person* is competent to perform a defined task — a welder, a non-destructive-testing inspector, an auditor.

WHY THE DISTINCTION HAS TEETH

A company can hold ISO 9001 (system) while selling an uncertified product, made by an uncertified welder. All three certificates are real, all three mean something different, and a customer who asks for the wrong one gets an answer that is true and useless.

What a certification scheme has to define

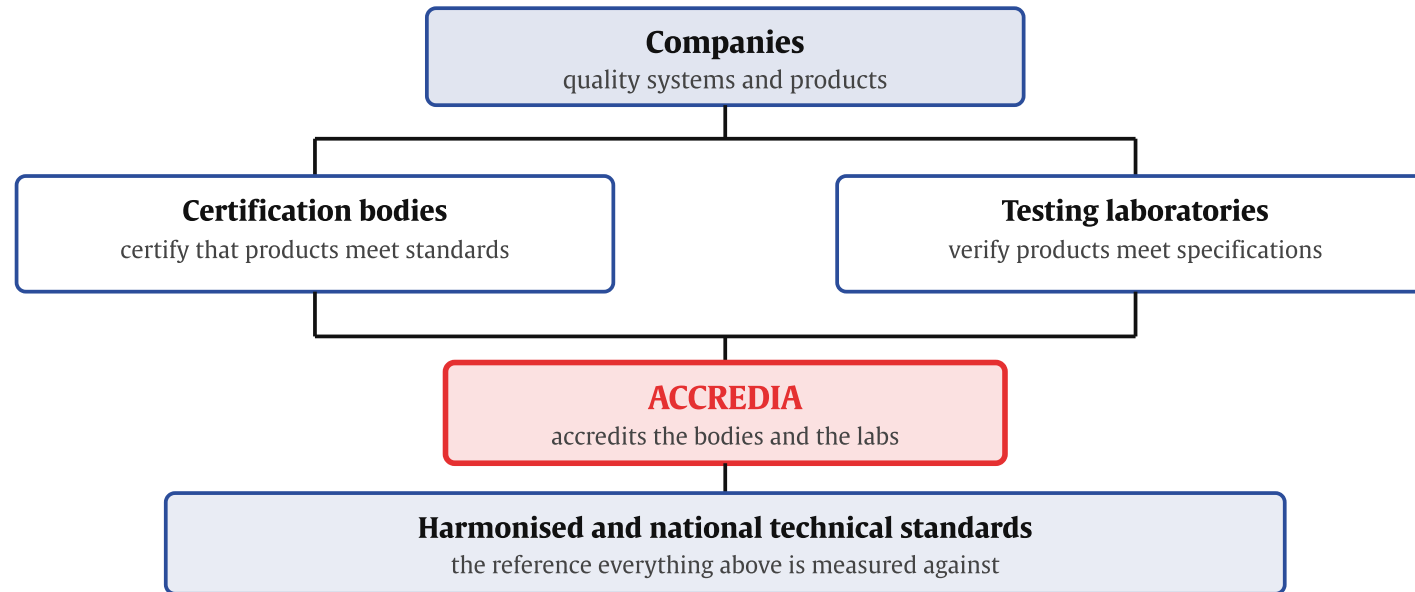
- **Product, service and process requirements** — the criteria the thing must meet.
- **Requirements for accrediting the scheme** — the conditions a certification body must satisfy to be recognised to issue certificates at all.
- **Testing methods for verification** — the procedures used to assess and confirm conformity.
- **Measurement, testing and calibration procedures** — how instruments are measured and calibrated so the results mean something.

Note where this course lives

The last item is the whole second half of this course. A certificate is only as good as the measurement behind it — which is why the rest of the semester is about measurement techniques.

Who does what

Four roles, and they must not be the same organisation



Why it has to be someone else

The independence argument, and its weak point

Each layer in that diagram exists because the layer below has a commercial interest in the answer.

The manufacturer

wants the product to pass

The certification body

checks the manufacturer — but is paid by them

The accreditation body

checks the certifier, and is not paid per certificate

THE UNCOMFORTABLE QUESTION

The certification body is chosen and paid by the company it audits. What stops it from being lenient in order to keep the contract? Think about it before the next slide — the answer is the reason accreditation exists at all.

THE STRUCTURAL ANSWER

An accredited body that issues easy certificates loses its accreditation, and with it every client who needed an accredited certificate. Its commercial incentive is therefore to be *strict enough to stay credible*. The system does not rely on virtue; it engineers the incentive.

Designated by the Italian government to assess the competence and impartiality of certification, inspection, verification and validation bodies, and of testing and calibration laboratories.

Three functions:

- **Accrediting bodies and laboratories** — checking they meet the requirements to do their work at all
- **Guaranteeing competence and impartiality** — that they are technically able, and have no stake in the outcome
- **Building confidence** in the results of conformity assessment

Accreditation is not certification

Certification says *your product conforms*. Accreditation says *the organisation that told you so is competent to say it*. One member state, one national accreditation body — that is Regulation (EC) 765/2008 from last time.

Product liability

Directive 85/374/EEC

The directive introduces **product liability**: the responsibility of those who place a product on the market if it causes damage.

The producer is **exempt** if they can demonstrate either of two things:

Compliance with a mandatory rule

The product was manufactured in accordance with a mandatory technical rule.

State of scientific knowledge

The knowledge available when the product was released did not allow the defect to be discovered.

WHY THIS CLOSES THE LOOP

Liability is what gives the whole system teeth. Standards are voluntary — but if your product injures someone and you cannot show you followed the state of the art, “it was not compulsory” is not a defence.



- **1 Standards are written by consensus bodies, not governments** — but law points at them and contracts rely on them, which makes voluntary documents effectively binding.
- **2 Certification is third-party.** The CE marking, by contrast, is the manufacturer's own first-party declaration.
- **3 Product, system and personnel certification are three different things.** Asking for the wrong one gets a true and useless answer.



- **1 Accreditation checks the certifier.** The system engineers the incentive rather than relying on virtue.
- **2 Liability makes all of it matter.** “It was not compulsory” is not a defence when someone is hurt.

Next time

The history of quality management — how we got from end-of-line inspection to quality systems, and why Japan changed the question.

Questions?

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